Toward a Finnish Volleyball Player Development Model

Working draft — outline, material, and decisions to make

A scaffold for a player-development framework for Suomen Lentopalloliitto, using the Swedish and Finnish basketball models (see `swedish-basketball-development-model.md` and `finnish-basketball-development-model.md`) as inspiration. **This is a draft to react to, not a finished framework.** Open decisions are flagged with **[DECIDE]** and collected at the end.

---

0. How to read this

- It's a skeleton: the sections, the spine, and the arguments — not yet the full text.
- It takes an explicit theoretical position (below) so we have something concrete to agree or argue with. Swap it out if you disagree; everything downstream follows from it.
- It is written to be honest about evidence and about the two basketball models' weaknesses, not to sell a method.
- **Show, don't just tell.** The theory (eco-D / nonlinear pedagogy) reads as abstract and academic, which is itself the biggest adoption risk (see §1). So the rule for the finished model is: **every principle is paired with a concrete, on-court volleyball example** — what the coach sets up, what the players do, what to look for. If a section can't be shown in practice, it doesn't belong. In this draft the examples are placeholders/sketches; the full version makes them the main event.

---

1. What the model is for — the problems and gaps it addresses

There isn't one single problem; there's a cluster the model should speak to. None of these is "the core" — they interact. **[DECIDE / EXPAND: which of these matter most to you, and what's missing.]**

- **No shared theory of how players learn.** Unlike Sweden's basketball framework, there is no published player-development framework grounded in a stated learning concept — so practice quality varies club to club and there's no common reference point.
- **A technique-first *tilt* — visible in the federation's own *lajilinjaukset*.** Credit where due: the published skill guidelines (*Lentopallo-/Lapsuusvaiheen-lajilinjaukset*) are framed as **Periaatteet + Opetusavaimet** (principles + teaching keys) — *not* a rigid "one correct technique" rulebook, and the principles framing is itself a move toward flexibility. **And they genuinely fold in contemporary ideas** — external-focus cues (*"pallo tottelee käsien kulmaa"*), game-reading (*"näe peli: pallo–passari–pallo–hyökkääjä"*), even one clear affordance line (*"hyökkääjällä on paljon tapoja tehdä pisteitä… tunnistaa tilanne ja etsiä keino tehdä piste"*). So the reservation is about **emphasis and coherence, not dogma**: the *first* principle on almost every page is still *"Yksinkertaiset liikkeet ovat tehokkaampia ja toistettavampia kuin monimutkaiset"* (simple movements are more effective and more repeatable than complex), and many teaching keys script footwork step-by-step — so the modern cues tend to **sit beside** an install-the-template default rather than **drive** the design. That's the same tension we flag for the basketball models (§3): a modern vocabulary layered over a one-true-technique habit (a milder cousin of the IP+eco-D blend Gray warns against, §4). The practical risk is a pull toward decontextualised, closed drills — especially for the youngest — that the skill-acquisition literature says transfer poorly (see the *ball-taming* and *messy-practice* fieldstones). And the assumption that tends to travel with this tilt — *that beginners must drill the decomposed "basics" before they can play* — **is not empirically supported**: head-to-head studies find no advantage (often a retention cost) for decomposing-to-basics vs learning the whole skill scaled down (§7). The better way to make things easier for a beginner is **task simplification, not decomposition**.
- **Coaches reproduce how they were coached.** Many coaches are ex-players who default to the methods they grew up with rather than choosing from evidence — the "copy-paste" problem the Swedish federation also named (Pleick's Lithuania story). One driver of the technique-first default, not the whole story.
- **The transfer gap.** Practice that looks tidy doesn't necessarily build players who read and decide in a real rally.
- **The theory sounds too theoretical.** Eco-D / nonlinear pedagogy comes wrapped in academic language (affordances, self-organisation, perception–action coupling) that makes practising coaches switch off. This is a real adoption barrier in its own right: a

model that can't be *shown on the court* won't be used, however correct. The antidote is relentless concreteness — see the "show, don't just tell" rule in §0.
- **A document alone changes nothing.** Whatever we write has to be paired with coach education, mentoring, and concrete tools, or it gets read and ignored (the implementation lesson from Basketball Immersion).

A useful sanity check for every later section: *does this actually help a coach run a better session next week?* — but that's a test, not the model's single mission.

---

2. What already exists (build on, don't duplicate)

Suomen Lentopalloliitto already has, as of the 2026 renewal:

- A **coach-education path** with a technique/skill foundation (**LTTV — Lentopallon tekniikka- ja taitovalmennus**) → age/role-banded tutkinnot: **Lasten lentopallovalmennus (U9–U13)**, **Juniorilentopallon valmentajatutkinto (U13–U17)**, **Lajivalmentajatutkinto (U17+)**, **Huippulentopallon valmentajatutkinto**, with **mentoring** built in.
- **Valmennuslinjaukset**: a *lajianalyysi* and a *fyysisen valmennuksen linjaus* (physical analysis), plus *lajilinjaukset*.
- An explicit statement that **coach development is non-linear** (good — it signals openness to the same idea on the player side).
- **Ecological dynamics is *already* in the federation's coach education.** The **Lasten lentopallovalmennus (LLV, U9–U13)** course explicitly covers *"miten mm. ekologisen dynamiikan periaatteita voidaan soveltaa käytännön valmennukseen"* (how to apply the principles of ecological dynamics in practical coaching). So the theory we're building on is **not foreign** — the federation has already endorsed it for the youngest age group.
- **Well-designed youth competition formats — already constrained, representative, and small-sided.** The mini-game ladder is genuinely modern: **U9 MiniLentis 2v2**, **U11/U13 3v3**, scaled nets/balls/courts, **Tiikeri/Super** difficulty tiers, and the 2025–26 additions — **Triplapallo (Triple Ball)** to guarantee three-touch rallies and **free reception** (fingers or forearm) — are essentially CLA / representative-design moves baked straight into the rulebook. **PowerCup** and **Junnulentis** are the participation vehicles (PowerCup mirrors the tiers as Tsemppi/Super). This is a major asset to build on — and a useful answer to anyone who thinks eco-D is foreign: *the game Finnish kids already play is constraint-led.*

The gap (= the opportunity) — an internal tension to resolve, not an import to sell: there is no published **player-development framework grounded in a stated learning theory** — the equivalent of Sweden's document. And there's a real tension *inside* the federation's own materials: **ecological dynamics in the U9–U13 coach course (LLV) and smart, constraint-led game formats (MiniLentis / Triplapallo), but a technique-template tilt in the written *lajilinjaukset*** (§1). The game design and the coach course already point one way; the skill guidelines and inherited coach habits point the other. That's the seam to work on. The model's pitch is therefore constructive: **carry the eco-D commitment the LLV already makes through into the skill guidelines and session design**, and reframe LTTV from "technique first" to "skill = adaptive solutions to the game." We are not replacing the coach-education path; we are giving it a coherent **theory of how players learn** that the federation has already started adopting.

---

3. What we take from the two basketball models

From Sweden (the strengths):

- A single, published framework with a **clear theoretical spine** (ecological dynamics / CLA), four pillars, and five "golden principles" for representative practice.
- The **game-model logic** (score by creating advantages; deny the opponent's) as the organiser of what skills matter.
- Constraints as the design tool; "keep it together," don't decompose.

From Finland basketball (the strengths):

- The **humanistic core**: ihmiskäsitys / oppimiskäsitys / pelikäsitys; the coach affects the game only *through the player*.
- **"US before ME"** and a team-first value base — a real fit for Finnish volleyball culture and a small talent pool.
- **Learning-to-learn and reflection** at the centre (self-assessment, dialogue; the Qridi-style reflection habit).
- **Konseptiajattelu**: teach the game's concepts together and revisit them with growing depth, instead of isolating one topic at a time.
- Socio-constructivist, **mentoring-based coach education** (which Lentopalloliitto already leans toward).

The cautionary lesson (don't repeat it): *both* basketball models, despite eco-D/pedagogy branding, **revert to information-processing** the moment they get concrete about decision-making (Sweden's SKANNA→TOLKA→BESLUTA→AKTION + "experience bank"; Finland's understand→read→decide). If we claim an ecological/representative foundation, we should **carry it through to how we talk about reading and deciding** , or be explicit and deliberate about being hybrid. Theoretical coherence is a feature.

---

4. Stance (decided)

The stance for this model: a **representative, constraint-led foundation grounded in ecological dynamics / CLA as the theory of skill learning — kept theoretically coherent (i.e. *not* blended with information processing)** — with **Finland's humanistic, team-first, reflection-driven coaching culture** layered on as the *coaching-relationship and learning-environment* dimension, and the whole thing **adapted to volleyball** and the Finnish system.

In one line: *"Älä irrota harjoitusta pelistä" — don't take practice out of the game* (ties to the existing follow-up-post idea in `fieldstones.md`).

Why "coherent, not hybrid" matters. Eco-D and information processing rest on **incommensurable assumptions** — direct perception / affordances / self-organisation vs internal models, representations, and computation. You can't coherently bolt them together; this is a point Rob Gray (ASU; *Perception & Action* podcast, *How We Learn to Move*) makes repeatedly, and it's exactly the trap *both* basketball models fell into by accident (eco-D branding, IP decision-making model). Because we've chosen eco-D, we **carry it through — including how we talk about reading and deciding**. No scan→interpret→decide→act pipelines, no "experience bank."

The Finnish layer is kept as pedagogy, not as a rival theory of skill. "Socio-constructivism" is itself a learning theory, and there's genuine philosophical tension between it and ecological dynamics (e.g. on the status of mental representation). So we are deliberate: the Finnish contribution lives at the level of **coaching relationships, values ("US before ME"), the learning environment, and reflection/learning-to-learn culture** — it does **not** get to describe how the motor skill itself forms (that's eco-D's job). Where the two would collide on a claim about cognition or representation, eco-D wins, and we phrase reflection in eco-D-compatible terms (attuning to information, education of attention) rather than "building internal models."

Alternatives considered and set aside:

- **(a) A theory-light LTAD/LTPD stage model** (Volleyball Canada/Ireland style) — familiar, fundable, agnostic about *how* skill is learned. Coherent because it mostly avoids the question. (We may still borrow its *stage scaffolding* — see §10 — just not as the theory.)
- **(b) An explicit IP / technique-first model** — i.e. formalise the status quo. Internally coherent, but it's the very approach the skill-acquisition evidence argues against for transfer.
- **(NOT recommended) A blended "IP + ecological" hybrid** — superficially "balanced," but theoretically incoherent (Gray), and it reproduces the basketball models' inconsistency. Flagged here only to rule it out deliberately rather than drift into it.

---

5. Proposed pillars (foundation)

Adapting Sweden's four + Finland's humanism to volleyball:

1. **The whole person (ihmiskäsitys).** The player before the position; warmth **and** quality.
2. **Development is non-linear.** Plan by developmental age, not birth year; maturation varies; no fixed ladder.
3. **Motivation, autonomy, and "US before ME."** Self-determination (autonomy/competence/relatedness) + the team-first Finnish value base.
4. **The coach designs the practice; the game does the teaching.** Learning is representative and **constraint-led**: the coach *designs* the environment (manipulating constraints) so that the information players read and the actions they take stay coupled, and so the thing practised stays the thing the match demands — letting players self-organise toward functional solutions. This is **not** "just let them play": the teacher is the **designed environment and the game's information** , not the coach's instructions *and* not unstructured play (the coach-as-*designer* role is the half the slogan "the game is the teacher" leaves out; see §7, and the "constraints are designed, not absent" caveat).

The conversation method for pillars 1 & 3: motivational interviewing (MI)

Pillars 1 and 3 need a concrete *how*, not just good intentions — and **motivational interviewing** is the method that operationalises them. MI (Miller & Rollnick) is a person-centred, autonomy-supportive conversation style for resolving ambivalence and supporting change; its "spirit" (partnership, acceptance, compassion, evocation) and core skills (OARS — open questions, affirmations, reflective listening, summaries) are almost a one-to-one expression of "the coach affects the game only through the player." MI and self-determination theory are tightly linked (Markland et al. 2005) — MI is essentially *how you support autonomy/competence/relatedness in conversation* — and OARS is the disciplined version of the "let the players reflect" principle (§8).

Kept in the pedagogy/relationship layer — no collision with the eco-D spine. MI is about motivation, goals, and the coach–athlete relationship, *not* a theory of how the motor skill forms. So it sits cleanly in the Finnish coaching-culture layer and never competes with ecological dynamics for the skill-acquisition job (per §4). It is for the *motivational/relational* conversation, not for verbal technical instruction of the movement.

Evidence, honestly: MI is a well-established, heavily-evidenced method (700+ trials; PA meta-analyses such as BMJ 2024 show MI-containing interventions raise activity), and there is now an authoritative sport-specific text — Rollnick, Fader, Breckon & Moyers, *Coaching Athletes to Be Their Best: Motivational Interviewing in Sports* (2019) — plus growing applied sport-psychology use. The honest caveat: sport-specific *performance-outcome* trials are still limited (Rollnick himself notes "MI is new to sports"), and even in health, isolating MI's specific added effect is harder than its popularity implies. We adopt it where its evidence is strongest — relationship, engagement, motivation, retention — not as a performance lever.

---

6. Why volleyball ≠ basketball (don't just port it)

Scope: the model is **indoor-first, with beach included where it shares the spine** (most of the foundation is shared; divergences are flagged, §6a). The two codes are not interchangeable for *representative design* — see below.

The model has to be built on volleyball's actual structure:

- **Rebound/rally sport — the ball may not rest.** This makes the *taming the ball* problem acute (see fieldstone): you can't dribble alone, so taming needs designed forms (wall, partner, varied self-feeds, unconventional contacts). Strong argument *against* the closed-drill default, *for* playful ball-mastery.
- **The net separates the teams.** No direct physical contest; the key **perception–action couplings span the net** — server↔passer, attacker↔block/defence, setter reading the block. Representative design must keep players reading *across the net*, which closed drills delete entirely.
- **Rotations and specialised roles** (setter, OH, MB, opposite, libero) — role literacy should **emerge from play**, not be prescribed early.
- **Skills are interception/timing-heavy** (serve-receive, set, attack, block, dig) and chained: **reception → set → attack** as one coupled sequence, not three separate techniques.
- **The rally as the unit.** The natural representative unit is a rally or a wash/segment, not a rep.

6a. Indoor vs beach: the representativeness asymmetry

A genuinely shared spine, but one structural difference changes practice design:

- **In beach, the small-sided game *is* the game.** 2v2 on a small court with modified rules already contains nearly all the real information, decisions, and contacts — so you get **high contact volume *and* high representativeness for free** just by playing with different rules/constraints. Representative design is comparatively easy.
- **Indoors, small-sided games diverge more from the whole 6v6 game** — different court width, more players, rotations, specialised roles, blocking structures. A 1v1 or 3v3 indoors is further from the real thing, so the indoor design challenge is **keeping small-sided/constrained games representative of the 6v6 whole** (e.g. preserving net width, role cues, the relevant reads) rather than assuming "small-sided = representative."
- **Design consequence:** beach can lean harder on near-full play; indoors needs more deliberate constraint design to bridge the gap between small-sided forms and the full game. This also nuances the §8 distribution — the "small-sided & constrained games" band carries *more representative load* in beach than indoors, where some of that work shifts to coupled-chain and modified-6v6 forms.

---

7. The skill-learning engine (CLA, carried through)

What we mean by *skill* (and why it isn't *technique*)

Before the engine, a definition — because the word "skill" hides a trap. There are **two different technique/skill distinctions** in circulation, and they are not compatible:

- **(A) The coaching-tradition one:** *technique* = the movement pattern; *skill* = choosing and applying "the right technique at the right time" (in the old shorthand, **skill = ability + technique**). This sounds sensible, but it is quietly **information-processing**: it assumes a *perceive → select the technique → execute* sequence, with execution separable from — and built before — the deciding. It is the frame the federation's *lajilinjaukset* still lean on (§1) and the one underneath the basketball models' decision layer (§3).
- **(B) The one we use (ecological dynamics):** *skill* = the **adaptive, functional relationship between player and game** — using the information in front of you to find and carry out a movement solution to *this* situation. *Technique* is then just **one coordination solution**, not a foundation to install first, and not separable from perceiving and deciding.

Working definition for the model (after Gray; Araújo & Davids): **skill is a player's ability to use information from the game to find and execute a movement solution** — an *attunement* to the game, not a stored template. Bernstein's image captures it: dexterity is "the ability to create a perfect key for any emerging lock." This is why §4 commits us all the way down: choosing eco-D means choosing definition **(B)**, so we deliberately avoid the seductive cliché "skill is the right technique at the right time," which smuggles the IP split back in. (Honest bound: this is *not* "technique work is forbidden" — an athlete's action is never wholly new, and bounded isolated work can widen the repertoire; the objection is to treating one technique as *the* solution or as a *prerequisite to playing* — the "organismic asymmetry" of over-investing in the athlete side at the expense of the task and environment.) Distilled sources in `finnish-volleyball-model-research-notes.md` §8.

- **Newell's constraints** — individual / task / environment — as the design vocabulary.
- **Representative learning design**: keep the information that's in the game (the net, the opponent, the read); don't decompose into on-air or coach-fed reps.
- **Constraint-manipulation logic** (adapted from Trinsey's volleyball CLA): (1) destabilise the existing solution, (2) encourage exploration/self-organisation, (3) amplify information / invite affordances (sometimes by making it *easier*), (4) give transition feedback about the search.
- **Repetition without repetition** — many varied contacts, never the same twice (cf. the basketball brief / fieldstones); this is how we keep volume *and* variability, answering the ex-player's reasonable worry that "they still need lots of contacts."
- **Decision-making as emergent**, tied to affordances across the net — *not* a scan→interpret→decide→act pipeline. This is where we deliberately avoid the basketball models' IP slip.
- **No one-true-technique**: functional movement solutions vary by body and situation; judge by the outcome. Directly counters the technique-first reproduction default.
- **What the volleyball perception research supports (carefully).** The volleyball expertise literature is mostly perceptual-cognitive (video/eye-tracking) and shouldn't be oversold as eco-D proof, but two findings transfer cleanly: (1) **representative/in-situ tasks reveal richer, different perception than decomposed lab tasks** — in-situ 6v6 defence shows skilled players use *more* exploratory visual search and read **functional spaces** (between attackers/blockers), and on-court vs off-court video yield different gaze behaviour (Piras/Castellar; JSSM 11(2):339; PMC10968438) — direct support for representative design; (2) skilled volleyball perception is about **reading information across the net** (server, setter, hitter, spaces), not running a closed motor pattern — support for "reads, not just reps."
- **"Task simplification, not decomposition" is the engine's rule for beginners.** To make a skill easier for a new learner, **scale the whole skill** (lighter/softer ball, lower net, smaller court, fewer players, slower feed) while keeping perception and action coupled — do *not* break it into closed parts. This is the core of Gray's *Advanced Ecological Approach* (2025) Ch. 7 ("the better alternative to a 'fundamentals first' approach").
- **The comparative evidence leans our way — and specifically kills "fundamentals first."** Across ~21 direct head-to-head training studies (Gray's running log; DL/CLA/NLP vs traditional, many sports), **when there's a difference it almost always favours the variability/ecological approach** — and almost none favour the traditional group; differences show up on the measures that matter for a game (adaptability, decisions, retention, creativity, even injury-risk markers). A sizeable minority show *no significant difference* — but even those usually add solution variety or retention for the eco group. Most directly: **decomposing to drill the "basics" first has no empirical advantage over learning the whole skill scaled down** (Deuker 2023, soccer — decomposed fundamentals vs small-sided games: no difference, better retention for games; Caldeira 2023, volleyball — CLA practice of the *spike against a block* produced more successful attacks than traditional/blocked practice).
- **Say the efficacy claim precisely — "equal technique, better decisions and transfer."** The 2025 *Sports Medicine – Open* review (Bromilow et al.) pins down the *shape* of the evidence, and the shape matters: **on technique/skill execution, NLP ≈ linear — parity, not superiority** (the game-based meta-analyses find *no significant difference* on skill execution: Abad Robles 2020; Ortiz 2023), while the advantage shows up on **decisions/tactics and transfer** (NLP favoured for tactical outcomes). So we **do not** claim NLP produces cleaner technique than drilling does; we claim it gets you the *same* technique **plus** the game-reading a rally actually needs — which is exactly why it's the

better bet for a *game*.

- **And that "parity" is probably an *understatement* — the test was rigged toward the drillers.** As Gray argues (2025, *The Advanced Ecological Approach to Skill Development*, Ch. 7), in most of these studies "technique" was measured with an **isolated, decontextualised** test — which is the traditional group's *home turf*: by specificity-of-practice you are best at exactly what you rehearsed, so a closed test should favour the group that drilled closed. That the eco-D/CLA group **matched (or beat) the drilling group even on that biased test** makes the result more striking, not less. The cleanest case is Deuker (2023): the outcome measure *was* the decomposed "fundamentals" test itself, yet the small-sided-games group still matched it **and retained better**. (The reviews flag the same blind spot from the other direction — outcomes are often assessed with instruments of low representativeness: Bromilow et al. 2025.) Had these studies tested in a *representative* way (in a real rally), the gap would likely widen in eco-D's favour. (Honesty check: this is the documented conclusion of the scoping pass in `finnish-volleyball-model-bibliography.md`, Appendix B — and the evidence base is real but thin and low-quality, so we lead with theory + best-available evidence, not "settled science.")
- **The honest counter-current.** It's *not* "eco always wins for everyone." Effects are **moderated by learner adaptability** (Yang 2025; the 2025 adaptability study — high-adaptability learners gain most) and possibly **skill level** (Chapman's cricket-spin thesis: CLA best for younger/less-skilled, traditional better for the *already-skilled*). And many studies are small, short, and methodologically weak. So we lead with the strong, defensible claim (no case for fundamentals-first; simplify don't decompose) and keep the humility.
- **Honest evidence caveats**: CLA is a framework, not a guarantee (Kennedy & O'Brien 2024 — coaches lack research-informed guidelines for choosing constraints); child-specific implicit-learning evidence is thinner than adult; "not a free-for-all" — constraints are designed, not absent.

---

8. Golden principles for session design (volleyball version)

Adapting Sweden's five "golden principles" to volleyball, written as checks an ex-player coach can apply on the spot:

1. **Many and varied contacts** — high volume, but vary it (location, ball, tempo, start). Repetition *without* repetition.
2. **Reads, not just reps** — is there something to perceive (a server, a block, a defender, the net)? If the decision is removed, it's not representative.
3. **Game-representative for the age** — right ball, right net height, right court, right information; scaled equipment is a *taming/representative* tool, not dumbing-down.
4. **Judge the solution, not the form** — did it solve the rally problem? Resist correcting toward one textbook technique.
5. **Let the players reflect** — short reflection/learning-to-learn loops (what did you notice / what worked), pair and small-group work.

(Plus a sixth volleyball-specific lens worth debating: **keep the chain intact** — practise reception→set→attack coupled rather than as isolated stations.)

What it looks like on the court (a worked example — the format every section should use)

This is the kind of concrete translation the finished model lives or dies on (§0). One example, to show the move from a closed drill to a representative game:

> **The closed drill (status quo):** coach tosses balls to a line of U12s; each player platform-passes the toss back to a target; coach corrects arm angle. No server, no decision, the ball arrives the same way every time, one contact then back of the line.
>
> **The representative version — "serve-receive to a target, 2 points if you can attack."** Pairs across a low net. One player **serves underhand** (real, varied information to read); receiver passes to a middle target zone; a third player sets; receiver tries to **attack/tip over**. Score: 1 point for a controlled pass, **2 points if the pass was good enough to attack off it**.
>
> - **Many & varied contacts:** every serve is different (location, speed) — varied reception reps, no two the same.
> - **Reads, not reps:** the receiver must read a live serve, not a coach toss.
> - **Game-representative:** the whole reception→set→attack chain is intact; the pass *means something* because it feeds an attack.
> - **Judge the solution:** did the pass let them attack? — not "were the arms at the textbook angle?"
> - **Constraint logic:** the *low net* + *underhand serve* makes success reachable

(invites the affordance); the *2-point bonus* steers attention to a passable, attackable ball without the coach saying "pass it high to the middle."
- **Reflect:** "which serves were hardest to pass, and what did you do differently?"

Same or greater contact volume than the line drill — but now it's reading, deciding, and the real chain. *That* is the pitch to a sceptical ex-player coach: not less work, just work that transfers.

This example is one worked case from the **Practice Design Guide** (`finnish-volleyball-practice-design-guide-DRAFT.md`) — the companion that turns these principles into a *design method* (a repeatable design loop + a smell-test + worked cases), deliberately **not a drill bank**, so coaches learn to design and adjust rather than copy.

Practice-time distribution (a deliberate cap on closed drilling)

Sweden gives coaches a practice-time "pie." We borrow the idea but **slice it by how representative the form is, not by which technique is trained** — because slicing by technique would quietly re-license the decompose-and-drill habit we're trying to change. The distribution is the single most concrete lever against the §1 "technique-first" default: it puts a visible ceiling on closed work.

Five bands, from most to least representative (indicative starting proportions — **[DECIDE: calibrate the numbers]**, and they shift by stage, §10):

Band	What it is	Younger (≈U9–U13)	Older (≈U16+)
Game & modified match play	6v6 and near-full rallies	~20%	~35%
Small-sided & constrained games	1v1–4v4, wash games, narrow-court, king/queen	~35%	~30%
Coupled chains with live information	reception→set→attack, serve↔pass, block↔attacker — kept whole, real reads	~20%	~25%
Ball-mastery / "taming" play	playful, varied self/partner/wall contacts; unconventional touches (see ball-taming fieldstone)	~20%	~5%
Isolated/closed technique	the old line-drill mode	**≤5%, and even this kept varied**	**≤5%**

The headline: even the *least* representative band is capped low, and at young ages its place is largely taken by **ball-mastery play**, not closed drilling. Physical/athletic development runs alongside this per the federation's existing *fyysinen lajianalyysi* and isn't counted in these bands.

---

9. Game model / peli-idea for volleyball [DECIDE on detail]

- **Core logic:** win the rally by manufacturing clean attacking chances and denying the opponent's, while managing error/risk — expressed through the **serve, serve-receive→attack, and transition** chains.
- **Principles of play over fixed systems early.** Like the Swedish "no predetermined decisions up to ~U16," up to age **[DECIDE]** players are free to break the pattern; role/rotation systems are introduced late and lightly.
- **Rotation/role literacy emerges from small-sided play** (1v1, 2v2, 3v3, queen/king of the court, narrow-court doubles, wash games) before full 6v6 specialisation.

---

10. Developmental stages (LTAD-based)

Decided: use a **full LTAD/LTPD-style developmental-age pathway** (à la Volleyball Canada/Ireland), from *liikkujasta urheilijaksi* / physical literacy through performance and *active for life*. It's a well-developed model, and it **doesn't collide with the existing Finnish system** — the federation's coach-education bands (U9–U13 / U13–U17 / U17+ / huippu) **map onto** the LTAD stages rather than competing with them, so we get the richer developmental detail *and* alignment with the coach pathway.

Volleyball is a *late-specialisation* sport (Sport for Life classification): the early stages are about broad movement and ball-mastery, not early position-locking. That fits the Finnish *monipuolisuus* ("Kasva urheilijaksi": *monipuoliset motoriset taidot ja fyysiset ominaisuudet / psyykkiset taidot ja elämäntaidot / lajivalmentautuminen*) and the Olympiakomitea phases (*lapsuus- / valinta- / huippuvaihe*).

The pathway (stages mapped across the three systems)

Ages are **indicative and developmental, not birth-year** — **[DECIDE: calibrate boundaries]**; and **[DECIDE]** how closely to localise the English stage labels vs the Finnish *liikkujasta-urheilijaksi* wording.

Stage (Sport for Life)	Finnish phase	Indicative age	Coach band
Active Start + FUNdamentals	lapsuusvaihe (varhainen)	~3–8	pre-LLV / koulu & seura
Learn to Train *(the skill-hungry window)*	lapsuusvaihe	~9–13	**LLV (U9–U13)**
Train to Train	valintavaihe (varhainen)	~13–16	**JLVT (U13–U17)**
Train to Compete	valintavaihe → huippuvaihe	~16–19+	**Lajivalmentaja (U17+)**
Learn to Win	huippuvaihe	~19+	**Huippuvalmentaja**
Active for Life	(rinnakkainen, mikä tahansa ikä)	any	harraste / beach

What each stage looks like (konseptiajattelu — the same concepts recur, deepened; they are not unlocked one at a time)

- **Active Start + FUNdamentals (~3–8).** Physical literacy and a *relationship with the ball* — throw/catch, balloon/beach-ball/soft-ball games, movement variety. **But this is emphatically *not* a "no net game" stage.** With the right constraints — light, slow ball; low net; small court; catches (*koppi*) allowed; movement with the ball permitted; no forced uncomfortable underhand passing — small children play the *real* game: Finland's **MiniLentis (U9/F) is a 2v2 net game** played in tournaments from roughly age 6–8. This is the model's own thesis in action (§7): you make the game reachable by **scaling and constraining it, not by decomposing it** into solo dribbling or catching drills. **A live design tension to flag, though:** Finland's MiniLentis keeps rallies alive by *allowing a catch* (*koppi*) — accessible, but the catch breaks the continuous perception–action coupling that makes the contact representative. **Petter Nilsson (SE) argues the other way: drop the catch altogether and keep the game *more* representative**, relying on the *other* scalings (lighter/slower ball, lower net, smaller court, no forced underhand pass) to keep it playable for the very young. Worth testing both — the catch buys participation; removing it buys representativeness. **[DECIDE: where we land on the catch.] Forms:** constrained 2v2 net play **and** movement/ball-taming games — the two coexist; taming is *not* a gate you must clear before you're allowed to play. **Distribution tilt (§8):** mostly play (constrained 2v2) + ball-mastery; **closed technique ≈0%. Not yet:** positions, technique flaw-hunting, full-size ball/net/court, 6v6.
- **Learn to Train (~9–13) — the window LLV and the *lapsuusvaiheen lajilinjaukset* govern.** The skill-hungry age: maximise varied contacts and reads through the federation's **scaled small-sided competition itself** — the ladder runs U9 **MiniLentis 2v2** → U11/U13 **3v3** (badminton-court width, nets 180/200 cm), each split into **Tiikeri (beginner) and Super (experienced)** rule levels so difficulty matches the player. The 2025–26 rule additions are textbook representative-design-by-rule: **Triplapallo (Triple Ball)** in the U11/U13 beginner series feeds the rally in three-ball cycles to *guarantee* reception→set→attack reps, and **free reception** (fingers *or* forearm) stops mandating one "correct" platform. **Forms:** 2v2–3v3 match play, modified rules, taming; **scaling:** lighter/softer balls, lower nets, smaller courts (a *representative* tool, not dumbing-down). **Distribution tilt:** small-sided games + ball-mastery dominate; closed ≤5% (and varied). **Not yet:** position specialisation, fixed 6v6 systems, technique flaw-hunting. *(This is where the §1 tension bites hardest — the **game formats already embody our principles**, yet the lapsuus* lajilinjaukset* and inherited coach habits still lean technique-template; closing that gap is the highest-leverage change.)*
- **Train to Train (~13–16).** Adolescence/growth: build the movement engine; introduce 6v6 *gradually* via **principles of play over fixed systems** (cf. Sweden's "no predetermined decisions up to ~U16", §9). Role literacy begins to *emerge*. Overlaps urheiluyläkoulu / "Kasva urheilijaksi." **Forms:** 6v6 introduced, heavy small-sided + coupled chains; full ball/net/court. **Distribution tilt:** shift toward game & coupled chains; ball-mastery shrinks. **Not yet:** locking a position too early, over-systematising.
- **Train to Compete (~16–19+).** Position specialisation and position-specific representative design; competition load and tactical concepts deepened. **Forms:** full 6v6, role-specific representative tasks. **Distribution tilt:** game + coupled chains dominate;

closed still ≤5%. **Not yet:** trading away adaptability/variability for early polish.
- **Learn to Win (~19+).** High performance: individualised constraint design, fine margins, full performance environment.
- **Active for Life (any age).** Lifelong play — recreational indoor, beach, *harrastelentopallo*. A parallel track with entry/exit at any point, not a failure state (the Sport for Life rectangle, not the old pyramid).

---

11. Coach education & implementation

This is where a framework lives or dies (the last bullet of §1): the document only matters if it changes what happens in the gym.

- **Reframe LTTV** from "technique first" to "skill = adapting to the game"; keep the contacts, change the context.
- **Give ex-players a bridge, not a rebuke.** Show that CLA delivers the volume and the "real coaching" they value — constraints that *guarantee* contacts and reads — so they don't experience it as "just letting them play." **Use an MI stance to change coaches, not just players:** mentors evoke a coach's *own* reasons to shift away from closed drilling rather than lecturing them — the same person-centred method (§5) applied to the §1 reproduction problem.
- **Lean on mentoring** (already in the 2026 model): change happens in the gym with a mentor, not in a lecture. Pair the framework with **video exemplars** of constrained sessions (the Swedish federation produced coach-education videos; we should too).
- **Implementation > document** (the Basketball Immersion lesson): clinics, a coach community, repeated exposure. A PDF alone changes nothing.
- **Competition design as a lever:** scoring/rules that reward game-representative behaviour; small-sided formats in **PowerCup / Junnulentis**; net/ball/court scaling mandated at young ages.
- **Make the default easy — but build designers, not recipe-followers.** Provide the **Practice Design Guide** (`finnish-volleyball-practice-design-guide-DRAFT.md`): a design loop + smell-test + worked design cases, explicitly *not* a drill bank. Teach the loop and the diagnostics first, present every example as "here's the thinking — now change it," so coaches can diagnose and adjust any task rather than just run ours. (Guards against "constraints as a bag of tricks.")

---

12. Open questions / decisions

- **[DECIDED] Theoretical stance** (§4): coherent eco-D/CLA as the theory of skill, with the Finnish humanistic/socio-constructivist contribution kept strictly as coaching pedagogy/culture (relationships, values, reflection) — never as a rival account of skill formation. Blended IP+eco "hybrid" ruled out as incoherent (cf. Rob Gray); eco-D carried through to reading/deciding (no IP pipelines).
- **[DECIDED] Scope: indoor-first, beach included where it fits the shared spine** (§6a). Most of the foundation is shared; divergences are flagged — notably the representativeness asymmetry (beach: small-sided ≈ the game, contacts + representativeness come easily; indoor: small-sided diverges more from 6v6, so it needs deliberate constraint design to stay representative). Sitting volleyball still open.
- **[DECIDED] Audience/format: layered** (as Sweden did) — one shared spine expressed at three depths: (1) a **coach-facing framework** (the full theory + principles + worked examples + distribution); (2) a **coach-education layer** that feeds the existing LTTV/tutkinto path, mentoring, and video exemplars; (3) a short **parent/club-facing** piece (the "why" — why practice looks messier/game-like and that's the point). The deeper layers stay coherent with the spine; the shallower ones translate, never contradict, it. (Still to decide: build order — likely coach-facing framework first.)
- **[DECIDED] Language: develop in English, deliver Finnish-first.** The end product is Finnish-first for its audience (with an English mirror likely), but the drafting/development phase stays in English — AI assistants' idiomatic Finnish is generally weak, so Finnish text is best produced as a deliberate, human-checked translation pass at the end rather than drafted natively. (Cf. the heavy hand-correction the Finnish coaching prose has already needed; see `.cursor/rules/finnish-coaching-prose.mdc`.) **Crucially, the established Finnish vocabulary already exists** (Kalaja/Jaakkola: *taito, rajoitteet, ei-lineaarinen pedagogiikka, representatiivinen oppimisympäristö, itseorganisoituminen, "näpit irti" -valmentaminen*) — the Finnish version must use these, **not invented terms** (research-notes §5).
- **[DECIDED] Stage structure** (§10): **full LTAD developmental-age pathway**, now built out as a concrete table mapping Sport for Life stages → Finnish phases (*lapsuus-/valinta-/huippuvaihe*) → federation coach bands (U9–U13/U13–U17/U17+/huippu), with per-stage forms, scaling, and the §8 distribution tilt — they don't collide. (Still open: exact age boundaries, and how closely to localise stage *labels* to the Finnish *liikkujasta-urheilijaksi* wording.)
- **[DECIDED] Relationship to the federation: starts as your own thinking piece** (from the perspective of a regional head coach), **built to mature toward federation adoption.**

Practical consequences: write with an honest, opinionated voice, but don't burn bridges; **build on and cite the federation's existing materials** (LTTV, valmennuslinjaukset, lajianalyysi, the 2026 coach-ed renewal) so it reads as a constructive evolution, not a teardown; keep it adoption-ready (the layered format, §12 above, supports this).

- **[DECIDED] How prescriptive:** principles **plus worked, on-court examples**, **plus a practice-time distribution** (Swedish-style, but built around *representativeness of form* rather than technique categories — see §8). The abstraction of eco-D/NLP only earns coach buy-in when shown in practice (§0 "show, don't just tell"), and a concrete distribution gives coaches a visible target.

---

13. Source material

The full, consolidated, deduplicated reference list lives in **`finnish-volleyball-model-bibliography.md`** — APA-style with one-line annotations and source-type tags (`[peer-reviewed]`, `[book]`, `[thesis]`, `[review]`, `[grey]`, `[org]`), and `[check: …]` flags on any metadata still to confirm. The deeper distillations and quotes behind each source are in `finnish-volleyball-model-research-notes.md`.

The bibliography is organised as: (1) Finnish federation & youth competition; (2) volleyball-specific CLA / representative design; (3) volleyball perception–action & expertise; (4) empirical CLA / motor-learning comparisons; (5) motor-learning theory — practice structure, variability & attentional focus (the cognitive counterpoint); (6) ecological-dynamics / CLA / NLP theory; (7) technique vs skill definitions; (8) game-based pedagogies (TGfU / Game Sense / Play Practice); (9) implicit learning, deliberate play & equipment scaling; (10) talent development, LTAD frameworks & physical literacy (incl. the critique); (11) Finnish-language pedagogy, coaching competence & athlete pathway; (12) coaching effectiveness, motivation & the whole-person layer (MI + SDT); (13) internal cross-references.

The load-bearing anchors (see the bibliography for full entries): the federation's own *lajilinjaukset* and the Junnulentis/PowerCup youth rules (§1, §2, §10); Moy, Renshaw & Gorman (2024) on serve exploration; Caldeira et al. (2023) and Deuker et al. (2023) plus Gray's comparative log and Gray (2025, Ch. 7) on simplification-not-decomposition (§7); Bromilow et al. (2025) and the moderation caveats (Chapman 2015; Yang et al. 2025); Bernstein, Newell, Araújo & Davids (2011) and Gray on the technique-vs-skill definition (§7 box); Renshaw et al. (2019), Chow et al. (2016) and Newell (1986) for the CLA/constraints spine; and Miller & Rollnick (2013) / Rollnick et al. (2019) for the MI layer (§5).